

On the Use of Herbal Medicines in Management of Inflammatory Bowel Diseases: A Systematic Review of Animal and Human Studies

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Abstract Because of potential adverse events and lack of effectiveness of standard therapies, the use of complementary and alternative medicines (CAM), particularly of herbal therapies, for inflammatory bowel disease (IBD) is increasing. Results from the use of herbal therapies for managing IBD are promising, and no serious adverse events have been reported from them. Herbal therapies show their benefit in managing IBD by different mechanisms such as immune system regulation, antioxidant activity, inhibition of leukotriene B₄, inhibition of nuclear factor-kappa B (NF- κ B), and antiplatelet activity. In this paper, all reported herbal therapies established in animal IBD models or used for managing human IBD are systematically reviewed and their possible mechanisms of action discussed. Conducting clinical trials with high quality and validity (randomized, double blinded, controlled, on a large number of patients) to obtain more conclusive results about the use of herbal therapies in IBD is recommended.

Keywords Inflammatory bowel disease · Herbal therapies · Mechanisms

Introduction

Inflammatory bowel disease (IBD) is characterized by chronic or relapsing immune system activation and inflammation within the gastrointestinal (GI) tract. Ulcerative colitis (UC) and Crohn's disease (CD) represent the two major forms of IBD [1]. These two forms are considered to be more frequent in Western countries and in areas with a high socioeconomic development but relatively infrequent in southern Europe. Sporadic reports have shown a lower incidence and a milder course of the disease in Mediterranean countries. Recent reports suggest an increase in both incidence and prevalence rates [2]. Because of potential adverse events and lack of effectiveness of standard therapies, the use of complementary and alternative medicines (CAM), particularly of herbal therapies, for chronic illness such as diabetes, osteoporosis, cancer, and liver diseases is widespread and increasing [3–6], and IBD is not excluded from this trend [7]. Different reasons were reported for the use of CAM by IBD patients, such as research of information about their disease and for optimum treatment, the wish to stop taking steroids, female gender, and the low level of confidence in their doctor [8, 9].

In a study evaluating the frequency of CAM usage in a French population with IBD, 21.2% of patients reported using CAM for their IBD [8]. In a large survey performed on IBD patients in Germany, 51.3% had experience with CAM, and significantly more UC patients (59.8%) than CD patients (48.3%) had experience with CAM. Homeopathy (52.9%) and herbal medicine (43.6%) were the most commonly used types of CAM [9]. In Canada, 2,847 IBD patients were asked whether they used CAM. Current or past use of CAM for IBD was reported by 47%, of whom 50% continued their use of CAM. Herbal therapies were the most commonly used (41% of CAM users).

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Improvements in sense of well-being, IBD symptoms, and sense of control over the disease were the most commonly reported benefits. Only 16% of prior CAM users reported any adverse effect of CAM use [10]. In research on the use of CAM by children and adolescents with IBD, 72% of the children were said by their parents to be using CAM, and among them, 24% was reported to use herbal therapies. The children who had never used any CAM therapies (nonusers) did not differ from the users in terms of gender, age, disease, or disease duration [11].

In this review, we encourage categorizing herbal therapies used to date for UC and CD, revealing their possible mechanisms of action.

Mechanisms of Action of Herbal Therapies

Despite many efforts, etiology and pathogenesis of IBD remain unclear. The most accepted hypothesis implicates a combination of one or more of the following factors: immune system dysregulation (caused by genetic or environmental factors), abnormal GI tract luminal factors, primarily dietary factors and the microorganisms constituting the GI tract flora, and defects in the GI mucosal barrier that allow luminal factors to penetrate into the mucosa [12, 13]. Herbal therapies have shown their effectiveness in managing IBD via several mechanisms discussed below.

Effects on Immune System

Some hypotheses propose that damage to the bowel mucosa occurs as a result of a dysregulated immune response to multiple mucosal antigens comprised within the constituents of the “normal” intestinal microbial flora. There is evidence for defective responses in both the innate and the adaptive immune systems in IBD [14]. The behavior of the cells mediating innate immunity such as neutrophil, macrophages, dendritic cells, and natural killer cells are altered in IBD. Defective mucosal helper T (Th) cell responses and greater expression of cytokines such as interleukin (IL)-1 β , IL-6, IL-12, tumor necrosis factor alpha (TNF- α), and interferon gamma (IFN- γ) were demonstrated in patients with IBD [15, 16]. Destruction of the intestinal mucosa in IBD is associated with amplified production of the mentioned cytokines. For example, TNF- α secretion leads to activation of further cytokine production by macrophages and in addition can act directly on intestinal epithelial cells, initiating the changes in epithelial permeability and ion transport that lead to mucosal inflammation and lesions [17]. Recent studies have confirmed the efficacy of anti-TNF- α compounds in managing both UC and CD [18–20]. These reports make an increasingly strong case for the role

of the immune system as being a key component or perhaps even the underlying trigger for IBD [15]. Thus, it seems that agents that can regulate T cells and proinflammatory cytokines have potential to reduce the inflammation score and thus ameliorate IBD. Recent studies have shown that some herbal therapies improve IBD in human or animal models in this way (Tables 1 and 2). Among them, most studies about curcumin from *Curcuma longa* demonstrate its potential to reduce proinflammatory cytokines such as IFN- γ , TNF- α , IL-1 β , and IL-12 [21–25]. A constituent from *Trypterigium wilfordii* [26] and chios mastic gum from *Pistacia lentiscus* [27] have reduced proinflammatory cytokines in patients with CD.

Antioxidant Potential

Oxidative stress is a potential etiological and/or triggering factor for IBD, because the damaging effects of reactive oxygen molecules (ROM) have been well established in the inflammatory process. Although some conflicting results exist, it seems that patients with IBD demonstrate excessive oxidized molecules compared with healthy control subjects in a variety of organic systems (e.g., GI tract, blood, and respiratory system) [1]. Recent studies have shown decreased total antioxidant capacity and increased ROM in patients with IBD [59, 60]. It was demonstrated that some herbal therapies can ameliorate IBD by their antioxidant activity. *Zataria multiflora* [29], curcumin from *C. longa* [22, 31, 32, 47], *Ziziphora clinopoides* [33], *Ginkgo biloba* [35, 37], essential oil of *Satureja khuzestanica* [34], glycoprotein from *Gardenia jasminoides* [38], ethanolic extract and nicotine from *Nicotiana tabacum* [42], and polysaccharide from *Rheum tanguticum* [61, 62] were shown to reduce indicators of lipid peroxidation, including thiobarbituric acid reactive substances (TBARS), myeloperoxidase (MPO), and/or malondialdehyde (MDA) in animal models of IBD. Some of them, such as *G. biloba*, glycoprotein from *G. jasminoides*, and polysaccharide from *R. tanguticum*, have improved antioxidant power by increasing superoxide dismutase (SOD), glutathion (GSH), and/or catalase (CAT). Detailed information about these studies are shown in Table 1. It has been demonstrated that total antioxidant power was increased by chios mastic gum from *P. lentiscus* in CD patients [27].

Inhibition of Leukotriene B4

Leukotriene B4, a metabolite of arachidonic acid, is a potent proinflammatory mediator playing a role in several inflammatory diseases such as rheumatoid arthritis, asthma, psoriasis, and IBD [63]. Overproduction of leukotriene B4

Table 1 Studies investigating herbal therapies in animal models of colitis

Study	Model	Species	Plant	Part	Results
Wang et al. [28]	Acetic acid	Mouse	<i>Polygonum multiflorum</i>	2,3,5,4'-Tetrahydroxy stilbene-2-O-beta-d-glucoside	↓Colon damage ↓MPO, MDA ↓NO level ↑SOD activity Better than 5-aminosalicylic acid (5-ASA)
Nakhai et al. [29]	Acetic acid	Mouse	<i>Zataria multiflora</i>	Total extract	↓MPO and TBARS ↓Macroscopic and microscopic scores
Hausmann et al. [30]	DSS	Mouse	<i>Plantago lanceolata</i>	Phenylethanoid acetoside	↓Histological score in acute colitis Amelioration of chronic colitis
Deguchi et al. [31]	DSS	Mouse	<i>Curcuma longa</i>	Curcumin	↓IFN-gamma secretion ↓Body weight loss ↓Histological colitis score ↓MPO ↓Disease activity index
Camacho et al. [32]	TNBS	Rat	<i>Curcuma longa</i>	Curcumin	↓NF-κb ↓MPO ↓TNF-α ↓Colonic nitrites levels & iNOS ↓COX-2
Ghafari et al. [33]	Acetic acid	Mouse	<i>Ziziphora clinopoides</i>	Total extract	↓MPO and TBARS ↓Macroscopic and microscopic scores
Ghazanfari et al. [34]	Acetic acid	Mouse	<i>Satureja khuzestanica</i>	Essential oil	↓Lipid peroxidation ↓MPO ↓Macroscopic and microscopic scores comparable to prednisolone
Zhou et al. [35]	TNBS	Rat	<i>Ginkgo biloba</i>	Extract	↓Severity of gross lesion score ↓Intensity of inflammatory response ↓Mucosa injury ↓Ulceration area Alleviation of colitis ↑SOD, ↓MDA, ↓TNF-α, NF-κbp65, and IL-6
Zhang et al. [36]	TNBS	Rat	<i>Curcuma longa</i>	Curcumin	↓Th1 cytokines (IL-12, IFN-gamma, TNF-α, IL-1) ↑Th2 cytokines (IL-4 and IL-10) ↑Proportion of IFN-gamma/IL-4
Zhang et al. [21]	TNBS	Rat	<i>Curcuma longa</i>	Curcumin	↓Macroscopic scores of colitis ↓COX2
Mustafa et al. [37]	Acetic acid	Rat	<i>Ginkgo biloba</i>	Extract	↓MPO ↓TNF-α, and IL-1β ↑GSH ↓Macroscopic colonic damage ↓Histopathological changes
Oh & Lim [38]	DSS	Mouse	<i>Gardenia jasminoides</i>	Glycoprotein	↓MPO, TBARS ↓NO, iNOS ↓COX-2 ↓NF-κbp50 ↓CAT, SOD, and GPx

Table 1 continued

Study	Model	Species	Plant	Part	Results
Cheon et al. [39]	DSS	Mouse	<i>Commiphora mukul</i>	Guggulsterone	↓NF- κ ↓ κ b Degradation ↓ κ b Kinase ↓Severity of colitis ↓Tissue upregulation of κ b & κ b kinase
Ukil et al. [40]	TNBS	Mouse	<i>Camellia sinensis</i>	Theaflavin-3,3'-digallate	↓TNF- α , IL12, IFN-gamma ↓iNOS ↓NF- κ b, κ ba ↓cytosolic κ b kinase
Jiang et al. [22]	TNBS	Rat	<i>Curcuma longa</i>	Curcumin	Improved survival rate & histological image ↓Macroscopic scores ↓MPO ↓COX-2 ↓Proinflammatory cytokines
Anthoni et al. [41]	DSS	Mouse	<i>Boswellia serrata</i>	Acetyl-11-keto-beta-boswellic acid	Antiinflammatory effect; protective responses were comparable to corticosteroid
Ko & Cho [42]	TNBS	Rat	<i>Nicotiana tobacum</i>	Ethanol extract of tobacco smoke, nicotine	↓Colonic lesion size ↓MPO ↓leukotriene B4 ↓IL-1 β
Kiela et al. [43]	DSS or TNBS	Mouse	<i>Boswellia serrata</i>	Ethanol extract	Ineffective in ameliorating colitis
Jian et al. [23]	TNBS	Rat	<i>Curcuma longa</i>	Curcumin	↑NF- κ b ↓Histopathologic signs of inflammation ↓NF- κ b ↓ κ b Degradation ↓Proinflammatory cytokine
Jian et al. [24]	TNBS	Rat	<i>Curcuma longa</i>	Curcumin	↓Histopathologic signs of inflammation ↓IL-1 β ↑IL-10
Kim et al. [44]	TNBS	Mouse	<i>Catalpa ovata</i>	Catalposide (CATP)	↓Weight loss & colonic damage & mucosal ulceration ↓TNF- α , IL-1 β , intra cellular adhesion molecu-1
Liu et al. [31]	TNBS	Rat	<i>Rheum tanguticum</i>	Polysaccharide	↓Ulcerative area and weight of colon ↓White blood cell ↑SOD ↓MPO
Liu et al. [29]	TNBS	Rat	<i>Rheum tanguticum</i>	Polysaccharide	↓Diarrhea, mortality, colon mass, ulcer areas ↓MPO ↓CD4 and T-cells
Murakami et al. [45]	DSS	Mouse	<i>Zingiber zerumbet</i>	Zerumbone (a sesquiterpenoid)	↓IL-1 β , TNF- α Suppression of colitis
Maity et al. [46]	TNBS	Mouse	<i>Camellia sinensis</i>	Thearubigin	↓Diarrhoea ↓Disruption of colonic architecture ↓Neutrophil infiltration ↓Lipid peroxidation ↓Serine protease activity ↓NO, iNOS ↓Th1 ↓NF- κ b

Table 1 continued

Study	Model	Species	Plant	Part	Results
Ukil et al. [47]	TNBS	Mouse	<i>Curcuma longa</i>	Curcumin	↓Diarrhea ↓Disruption of colonic architecture ↓Neutrophil infiltration ↓Lipid peroxidation ↓MPO ↓MDA ↓Serine protease activity ↓Th1 and Th2 cytokines ↓NF- κ b ↓NO, iNOS
Micallef et al. [48]	DSS	Mice	<i>Polygonum tinctorium</i>	Tryptanthrin	↓Colon damage ↓IL-2 and IFN-gamma
Sugimoto et al. [25]	TNBS	Mouse	<i>Curcuma longa</i>	Curcumin	↓CD4 and T-cell infiltration ↓NF- κ b activation ↓Proinflammatory cytokine Improvement of wasting & histologic signs of inflammation
Kriegelstein et al. [49]	Indomethacin	Rat	<i>Boswellia serrata</i>	Boswellia extract and acetyl-11-keto-beta-boswellic acid	↓Rolling and adherent leukocytes ↓Tissue injury score ↓Macroscopic and microcirculatory inflammatory feature

TNBS 2,4,6-Trinitrobenzene sulfonic acid, *DNBS* 2,4-dinitrobenzene sulfonic acid, *DSS* dextran sodium sulfate, *SOD* superoxide dismutase, *MDA* malondialdehyde, *TNF* tumor necrosis factor, *NF* nuclear factor, *IL* interleukin, *MPO* myeloperoxidase, *GSH* reduced glutathione, *TBARS* thiobarbituric acid reactive substances, *NO* nitric oxide, *NOS* nitric oxide synthase, *COX* cyclooxygenase, *CAT* catalase, *GPx* glutathione peroxidase, *MCP* monocyte chemoattractant protein, *PG* prostaglandin, *LPS* lipopolysaccharide

was seen in IBD [64]. Thus, it appears that inhibition of this mediator can reduce inflammation and help to improve IBD. Nicotine from *N. tabacum* has been shown to reduce leukotriene B4 concentration in dinitrobenzene-sulfonic-acid-induced colitis [42].

Antimicrobial Effects

Some studies have suggested the role of microbial content of the GI tract in the pathogenesis of IBD. The disease occurs in areas of the GI tract with the highest concentrations of luminal bacteria. The complex enteric immune system regulates interactions between microbial antigens and immunocompetent cells. The inherited barrier function of the intestinal mucosa prevents bacterial translocation through, and invasion into, the bowel wall and a subsequent pathological tissue reaction. Normal, nonpathogenic enteric bacteria induce chronic intestinal inflammation in genetically susceptible hosts with defective immunoregulation, bacterial clearance, or mucosal barrier function [65, 66]. It has been shown that the concentration of intestinal bacteria in IBD is higher than normal and increases progressively with disease severity [67]. Therefore, modification of the

gut bacterial flora by antibacterials may be effective in treating UC [68, 69]. We found no study about antibacterial herbal therapies on the luminal bacterial content and disease improvement in animal or human models of IBD.

Nuclear Factor-Kappa B

Nuclear factor-kappa B (NF- κ B) proteins are a family of structurally related eukaryotic transcription factors that promote the expression of more than 150 genes, many of which play important roles in the regulation of inflammation and programmed cell death (apoptosis) [70, 71]. Excess or inappropriate activation of NF- κ B has been observed in human IBD [72, 73]. Although the activation of NF- κ B is not causative for inflammatory bowel disease, NF- κ B seems to play a major role in inflammatory responses in these diseases [70]. The NF- κ B proteins are associated with a family of inhibitory proteins known as I κ B. One of the key steps in activating the NF- κ B pathway is the stimulation of the I κ B kinase (IKK) that inactivates I κ B by phosphorylation. The IKK is essential in transducing the signal-inducible activation of the transcriptional factor NF- κ B in response to proinflammatory stimuli [74].

Thus, the inhibitors of NF- κ B or IKK can be used as a treatment strategy for managing IBD. Among herbal therapies, curcumin from *C. longa* [23], a plant sterol from *Commiphora mukul* [39], and theaflavin-3,3'-digallate from *Camellia sinensis* [40] may reduce NF- κ B and IKK and increase I κ B. Investigating the efficacy of ethanolic extract of *Boswellia seratia* in animal models of colitis showed that this plant causes an increase in NF- κ B and is ineffective in ameliorating colitis [43].

Effects on Nitric Oxide and Inducible Isoform of Nitric Oxide Synthase

IBD have been found to be associated with an overproduction of nitric oxide (NO) by the inducible isoform of NO synthase (iNOS) [75]. Increased luminal and salivary NO have also been detected in IBD patients [59, 76, 77]. Herbal therapies such as a glucoside from *Polygonum multiflorum* [28], curcumin from *C. longa* [32], a glycoprotein from *G. jasminoides* [38], and theaflavin-3,3'-digallate from *C. sinensis* [40] have demonstrated their efficacy in IBD by reducing NO and iNOS levels (Table 1).

Effects on Cyclooxygenase (Cox)

There are two known isoforms of cyclooxygenase: Cox-1 and Cox-2. They catalyze the synthesis of prostaglandins. Cox-1 is a constitutive enzyme, and prostaglandins produced by Cox-1 are thought to play a major role in the maintenance of GI homeostasis, including gastric cytoprotection and blood flow. Cox-2 is induced by exposure of transformed colonic cell lines to proinflammatory cytokines including IL-1 and TNF- α . Prostaglandins synthesized through Cox-2 mediate the inflammatory response [78, 79]. Different studies reported that curcumin from *C. longa* reduced Cox-2 level [22, 32, 36]. Another herbal therapy with this effect is *G. jasminoides* [38].

Antiplatelet Activity

Patients suffering from exacerbations of IBD have an increase in circulating platelet numbers [80, 81]. Recent work indicates that platelets exhibit several proinflammatory properties, including release of inflammatory mediators, and recruitment, chemotaxis, and modulation of the activity of other inflammatory cells [20, 82]. Thus, antiplatelet drugs may provide new therapeutic possibilities in managing IBD [83]. It was demonstrated that *Angelica sinensis* injection can significantly inhibit platelet activation, relieve vascular

endothelial cell injury, and improve microcirculation in patients with UC [52].

Clinical Trials

There are several clinical trials on the use of herbal therapies for patients with IBD, and the results obtained from these trials are encouraging. It should be remembered that randomized double-blind trials with a larger number of patients have higher quality and the results obtained from them are more valid. The results obtained from some of clinical trials on the efficacy of herbal remedies in IBD are mentioned below.

Clinical remission, clinical improvement, and clinical response were seen in significantly more UC patients using *Aloe vera* gel compared with placebo group. Clinical colitis activity index and histological scores were significantly reduced in these patients [51]. Administration of capsules containing *Artemisia absinthium* to CD patients led to complete remission of symptoms in 65% compared with none in the placebo group [50]. Grass juice from *Triticum aestivum* reduced overall disease activity index and severity of rectal bleeding in UC patients [53]. Table 2 shows detailed data about clinical trials on the use of herbal therapies in IBD.

Conclusion

Because of a lack of safety and efficacy of standard therapies, the use of herbal therapies for managing IBD is increasing, and such therapies have shown promising results. Herbal therapies have decreased the disease activity index, increased the rate of clinical response, shown clinical improvement and clinical remission, decreased the relapse rate, improved rectal bleeding and stool consistency, and reduced histological damage in patients with IBD. According to the role of immune dysregulation, oxidative stress, leukotriene B₄, NF- κ B, platelets, and iNOS in the pathogenesis of IBD, herbal therapies show their effectiveness by several mechanisms, such as immune system regulation, antioxidant activity, inhibition of leukotriene B₄, inhibition of NF- κ B or IKK, antiplatelet activity, and inhibition of iNOS. Unlike the potential role of microbial flora of GI the tract in the pathogenesis of IBD, there were no studies on the effects of antibacterial herbal therapies on the luminal bacterial content and disease improvement in animal models or humans. There are some difficulties in investigating the efficacy of herbal remedies in IBD. As concluded by the study of Langmead and Rampton [7], lack of reproducibility and directions for use of herbal products, nonadherence to standard

Table 2 Clinical trials investigating the efficacy of herbal therapies in patients with inflammatory bowel disease (IBD)

Study	Study design	No. of patients	Type of IBD	Plant comparator	Comparator	Part of plant	Duration of treatment	Results
Omer et al. [50]	Randomized, double-blind placebo-controlled	40	CD	<i>Artemisia absinthium</i>	Placebo	Powder	10 weeks	65% with complete remission of symptoms in plant group compared with none in the placebo group
Ren et al. [26]	Open, prospective	20	CD	<i>Tripterygium wilfordii</i>	–	T2 (major constituent of extract)	12 weeks	Significant ↓ in serum levels of CRP, TNF- α , IL-1 β , ↓ CDAI scores, endoscopic improvement
Kaliora et al. [27]	Open, comparing CD patients with healthy volunteers	18	CD	<i>Pistacia lentiscus</i>	–	Gum	4 weeks	Significant ↓ CDAI, ↓ IL-6, ↓ CRP, and ↑ TAP as compared to pretreatment values, not any side effects reported
Langmead et al. [51]	Randomized, double-blind, placebo-controlled	44	UC	<i>Aloe vera</i>	Placebo	Gel	4 weeks	Clinical remission, improvement and response in 9, 11, 14 patients of aloe group respectively compared with 1, 1, 2 patients of placebo respectively, significant ↓ of clinical colitis activity index and histological scores in aloe group
Dong et al. [52]	Randomized, uncontrolled	64	UC	<i>Angelica sinensis</i>	Routine therapy	Extract	3 weeks	No significant change in sigmoidoscopic score Significant ↑ platelet activation, relief of vascular endothelial cell injury, and improvement of microcirculation by herbal therapy compared with routine therapy
Ben-Arye et al. [53]	Randomized, double-blind, placebo-controlled	23	UC	<i>Triticum aestivum</i>	Placebo	Grass juice	4 weeks	Significant ↓ overall disease activity index, and ↓ severity of rectal bleeding, not reported any serious side effects
Gupta et al. [54]	Controlled	30	UC	<i>Boswellia serrata</i>	Sulfasalazine	Gum resin	6 weeks	Comparable efficacy with sulfasalazine in remission rate, improvement of stool properties and histopathology
Gerhardt et al. [55]	Randomized, double-blind	102	CD	<i>Boswellia serrata</i>	Mesalazine	Extract	8 weeks	↓ CDAI with Boswellia was comparable with mesalazine
Fernández-Bañares et al. [56]	Open label, parallel-group, randomized	105	UC	<i>Plantago ovata</i>	Mesalazine	Seed	12 month	As effective as mesalazine in maintenance of remission
Gupta et al. [57]	Controlled	ND	UC	<i>Boswellia serrata</i>	Sulfasalazine	Gum resin	6 weeks	Comparable remission rate with sulfasalazine
Greenfield et al. [58]	Randomized, placebo-controlled	43	UC	<i>Oenothera biennis</i>	Omega-3 fish oil, olive oil	Evening primrose oil	6 month	Significant improvement of stool consistency compared with omega-3 and olive oil, no significant change in stool frequency, rectal bleeding, disease relapse, sigmoidoscopic appearance or rectal histology in 3 treatment groups

IBD inflammatory bowel disease, UC ulcerative colitis, CD Crohn's disease, CRP C-reactive protein, TNF tumor necrosis factor, IL interleukin, TAP total antioxidant power, CDAI Crohn's disease activity index, ND not determined

requirements of licensing, inability to exactly specify the intervention, and difficulty finding published trials in languages other than English are the main problems. Therefore, we suggest conducting more clinical trials, with high quality and validity, to obtain convincing results on the use of herbal therapies in IBD.

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